

Applications of Derivatives — Revision Summary

Monotonicity, extrema, geometry, algebra, inequalities, and bounds

1 Monotonicity — increasing and decreasing functions

Theorem (Monotonicity test). Let f be continuous on $[a, b]$ and differentiable on (a, b) .

- If $f'(x) \geq 0$ on (a, b) , then f is non-decreasing on $[a, b]$.
- If $f'(x) > 0$ on (a, b) , then f is strictly increasing on $[a, b]$.
- Same with ≤ 0 and < 0 for non-increasing / strictly decreasing.
- If $f'(x) = 0$ on (a, b) , then f is constant on $[a, b]$.

Subtle point. Strict positivity of f' on the open interval is sufficient for strict monotonicity on the closed interval, by MVT. Conversely, strict monotonicity does *not* require $f' > 0$ everywhere: $f(x) = x^3$ is strictly increasing on $\mathbb{R}$ but $f'(0) = 0$. So the converse is “ f strictly increasing $\Rightarrow f' \geq 0$,” not $f' > 0$.

Standard recipe. To show $f \leq g$ on an interval, define $h = g - f$, show $h(a) = 0$, and show $h' \geq 0$. This collapses an inequality between values into one about a derivative.

Example. Show $\sin x \leq x$ for $x \geq 0$. Let $h(x) = x - \sin x$. Then $h(0) = 0$ and $h'(x) = 1 - \cos x \geq 0$, so h is non-decreasing on $[0, \infty)$, giving $h(x) \geq 0$ as required.

1.1 Critical and stationary points

A *critical point* of f is any point where $f' = 0$ or f' does not exist (within the domain). A *stationary point* is specifically where $f' = 0$. Maxima/minima of differentiable functions occur at stationary points (by Fermat), but not every stationary point is an extremum (e.g., inflection at $x = 0$ for $f(x) = x^3$).

2 Maxima and minima

2.1 Local extrema

Theorem (Fermat’s stationary-point theorem). If f has a local extremum at an interior point c and $f'(c)$ exists, then $f'(c) = 0$.

First-derivative test. Suppose c is a critical point and f is continuous at c .

- f' changes from $+$ to $-$ at $c \Rightarrow$ local **maximum**.
- f' changes from $-$ to $+$ at $c \Rightarrow$ local **minimum**.
- No sign change $\Rightarrow$ neither (often an inflection point).

Second-derivative test. Suppose $f'(c) = 0$ and $f''(c)$ exists.

- $f''(c) > 0 \Rightarrow$ local minimum.
- $f''(c) < 0 \Rightarrow$ local maximum.
- $f''(c) = 0 \Rightarrow$ inconclusive — check higher derivatives or use first-derivative test.

n th-derivative test (when second is inconclusive). If $f'(c) = f''(c) = \dots = f^{(n-1)}(c) = 0$ and $f^{(n)}(c) \neq 0$:

- n even, $f^{(n)}(c) > 0$: local minimum; $f^{(n)}(c) < 0$: local maximum.
- n odd: neither — it’s an inflection (with horizontal tangent if $n \geq 3$).

2.2 Global extrema on closed intervals

To find the absolute maximum/minimum of f on $[a, b]$ (where f is continuous): evaluate f at all critical points in (a, b) and at the endpoints a, b . The largest and smallest values are the global extrema. (Existence is guaranteed by the Extreme Value Theorem.)

On open or unbounded intervals. EVT may fail. Check behaviour at endpoints (or as $x \rightarrow \pm\infty$) to determine whether extrema are attained. Standard examples: $f(x) = x$ on $(0, 1)$ has no max or min; $f(x) = e^{-x}$ on $[0, \infty)$ attains its max at 0 but no min (infimum 0 is not attained).

2.3 Standard optimization problems

- **Largest rectangle inscribed in a circle/ellipse:** parametrise, write area as a function of the parameter, differentiate.
- **Minimum-surface-area cylinder of fixed volume:** $V = \pi r^2 h$ fixed, minimise $S = 2\pi r^2 + 2\pi r h$. Optimum: $h = 2r$.
- **Snell’s law from Fermat’s principle of least time:** differentiate travel time, get the law of refraction.
- **Distance-from-a-point to a curve:** minimise d^2 rather than d to avoid square roots.

3 Rate of change

The derivative $\frac{dy}{dx}$ is the instantaneous rate of change of y with respect to x . Two threads:

3.1 Related rates

If two variables x, y depend on time t and are linked by a relation $F(x, y) = 0$, differentiate implicitly with respect to t :

$$F_x \cdot \frac{dx}{dt} + F_y \cdot \frac{dy}{dt} = 0.$$

Standard problems. A ladder of length ℓ slides down a wall: $x^2 + y^2 = \ell^2$ gives $x\dot{x} + y\dot{y} = 0$, hence $\dot{y} = -(x/y)\dot{x}$. A spherical balloon expands: $V = \frac{4}{3}\pi r^3$, so $\dot{V} = 4\pi r^2 \dot{r}$.

3.2 Linear approximation and differentials

Near $x = a$:

$$f(x) \approx f(a) + f'(a)(x - a).$$

Equivalently, $df = f'(x)dx$. Used to estimate small changes: $\sqrt{4.02} \approx 2 + \frac{1}{4}(0.02) = 2.005$.

Error propagation. If $y = f(x)$ and x is measured with error Δx , then $\Delta y \approx f'(x)\Delta x$, and the relative error is $\frac{\Delta y}{y} \approx \frac{f'(x)}{f(x)}\Delta x$ — the *logarithmic derivative* times Δx .

4 Tangents and normals

For $y = f(x)$, the tangent and normal at $(a, f(a))$ are:

$$\text{Tangent: } y - f(a) = f'(a)(x - a), \quad \text{Normal: } y - f(a) = -\frac{1}{f'(a)}(x - a) \quad (f'(a) \neq 0).$$

If $f'(a) = 0$: tangent is horizontal, normal is vertical $x = a$.

If $f'(a)$ is undefined (e.g., infinite): tangent is vertical, normal is horizontal.

4.1 Tangent at parametric and implicit curves

- **Parametric:** $x = x(t), y = y(t)$. Slope = $\frac{dy/dt}{dx/dt} = \frac{\dot{y}}{\dot{x}}$ (when $\dot{x} \neq 0$).
- **Implicit:** $F(x, y) = 0$. Slope = $-\frac{F_x}{F_y}$ (when $F_y \neq 0$).
- **Polar:** $r = r(\theta)$. Convert to $(x, y) = (r \cos \theta, r \sin \theta)$ and use parametric formula.

4.2 Angle between two curves

The angle between two curves at a point of intersection is the angle between their tangents:

$$\tan \phi = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|.$$

Two curves are *orthogonal* at the intersection iff $m_1 m_2 = -1$ (or one is vertical and the other horizontal).

4.3 Newton's method

To approximate a root of $f(x) = 0$, iterate the tangent-line update:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

Geometrically, follow the tangent at $(x_n, f(x_n))$ to where it crosses the x -axis. Quadratic convergence near a simple root — the number of correct digits roughly doubles each step.

5 The log-derivative technique in algebra

For a polynomial $p(x) = a_n \prod_{i=1}^n (x - r_i)$ with roots $r_1, \dots, r_n$ (counted with multiplicity), the *logarithmic derivative* is

$$\frac{p'(x)}{p(x)} = \sum_{i=1}^n \frac{1}{x - r_i}.$$

Differentiating $\log p$ converts a product into a sum — the algebraic key.

5.1 Reciprocal power sums of roots

Let $S_k = \sum_{i=1}^n \frac{1}{r_i^k}$ be the k th reciprocal power sum (assume all $r_i \neq 0$).

First reciprocal sum. Set $x = 0$ in $\frac{p'(x)}{p(x)} = \sum \frac{1}{x - r_i}$:

$$\frac{p'(0)}{p(0)} = \sum_{i=1}^n \frac{1}{-r_i} = -S_1, \quad \text{so} \quad S_1 = -\frac{p'(0)}{p(0)}.$$

Higher reciprocal sums. Differentiate $\frac{p'(x)}{p(x)}$ once more:

$$\frac{d}{dx} \left(\frac{p'}{p} \right) = - \sum_{i=1}^n \frac{1}{(x - r_i)^2}.$$

At $x = 0$: $S_2 = \sum \frac{1}{r_i^2}$ comes out from this expression. Iterating gives $S_3, S_4, \dots$ in terms of p and its derivatives at 0.

Alternative via Vieta. If $p(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$, the elementary symmetric functions of $1/r_i$ correspond to coefficients of the *reverse* polynomial $\hat{p}(x) = x^n p(1/x) = a_0 x^n + a_1 x^{n-1} + \dots + a_n$. The roots of $\hat{p}$ are $1/r_i$. Vieta on $\hat{p}$ then gives:

$$S_1 = \sum \frac{1}{r_i} = -\frac{a_1}{a_0}, \quad e_2(1/r_i) = \sum_{i < j} \frac{1}{r_i r_j} = \frac{a_2}{a_0}, \quad \dots$$

Power sums S_k then come from Newton's identities applied to $\hat{p}$.

Worked example. Find $\sum 1/r_i$ and $\sum 1/r_i^2$ for the roots of $x^3 - 6x^2 + 11x - 6 = 0$. Here $p(0) = -6$, $p'(x) = 3x^2 - 12x + 11$, so $p'(0) = 11$, giving $S_1 = -11/(-6) = 11/6$. (Check: roots are 1, 2, 3, so $S_1 = 1 + \frac{1}{2} + \frac{1}{3} = \frac{11}{6}$.) For S_2 : by Newton's identity, $S_2 = S_1 e'_1 - 2e'_2$ where e'_k are elementary symmetric in $1/r_i$. From $\hat{p}(x) = -6x^3 + 11x^2 - 6x + 1$: $e'_1 = 11/6$, $e'_2 = 6/6 = 1$, so $S_2 = (11/6)^2 - 2 = 49/36$. (Check: $1 + \frac{1}{4} + \frac{1}{9} = \frac{49}{36}$.)

6 Establishing inequalities via derivatives

The most powerful general method for proving inequalities $f(x) \geq g(x)$ on an interval I :

Standard recipe. Define $h(x) = f(x) - g(x)$. Show:

1. $h(a) = 0$ at some convenient endpoint $a \in I$ (or $\lim_{x \rightarrow a} h = 0$).
2. $h'(x) \geq 0$ on $I \cap [a, \infty)$ — so h is non-decreasing from 0.

Conclude $h(x) \geq 0$, i.e., $f(x) \geq g(x)$, on the relevant region.

Refinement: cascade of derivatives. If h' is also hard to sign, repeat: show $h''(x) \geq 0$ and $h'(a) = 0$ to conclude $h'(x) \geq 0$, then $h(a) = 0 \Rightarrow h(x) \geq 0$. Iterate as needed — each layer trades the inequality for one about a higher derivative.

6.1 Convexity-based inequalities

If f is convex on I ($f'' \geq 0$), the chord lies above the graph and the tangent lies below:

$$f(\lambda a + (1 - \lambda)b) \leq \lambda f(a) + (1 - \lambda)f(b) \quad (\text{chord above}),$$

$$f(x) \geq f(a) + f'(a)(x - a) \quad (\text{tangent below}).$$

Tangent-line family. A library of useful one-line inequalities:

- $e^x \geq 1 + x$, with equality iff $x = 0$ (tangent to e^x at 0).
- $\ln(1 + x) \leq x$ for $x > -1$, equality at 0 (tangent to $\ln$).
- $\ln(1 + x) \geq \frac{x}{1 + x}$ for $x > -1$ (tangent to $\ln(1 + x)$ at 0 from the secant viewpoint).
- $\sin x \leq x$ for $x \geq 0$; $\sin x \geq x - \frac{x^3}{6}$ (Taylor remainder, alternating series).
- $\cos x \geq 1 - \frac{x^2}{2}$ for all x .
- AM-GM: $\frac{a_1 + \dots + a_n}{n} \geq \sqrt[n]{a_1 \dots a_n}$, from concavity of $\ln$ via Jensen.

6.2 Strange-looking inequalities, demystified

Symmetry and convexity. An inequality involving $f(x_1), \dots, f(x_n)$ with a constraint $\sum x_i = \text{const}$ often yields to Jensen's inequality. The right move: identify the convex/concave function in disguise.

Example. Prove $\frac{x^x \cdot y^y}{x \cdot y} \geq \left(\frac{x+y}{2}\right)^{x+y} / (xy)^{(x+y)/2} \dots$ etc. Take logs, rephrase as a Jensen statement about $f(t) = t \log t$ (convex on $(0, \infty)$).

Schur-like inequalities. Often handled by reducing to a single-variable optimisation: fix the sum or product of variables, vary one, and study the resulting one-variable function.

7 Bounds via Taylor expansion

Theorem (Taylor's theorem with Lagrange remainder). If f is $(n+1)$ -times differentiable on $[a, x]$:

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots + \frac{f^{(n)}(a)}{n!}(x-a)^n + R_n(x),$$

where $R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-a)^{n+1}$ for some ξ between a and x .

Bounding R_n . If $|f^{(n+1)}| \leq M$ on the interval, then $|R_n(x)| \leq \frac{M}{(n+1)!}|x-a|^{n+1}$. This is how Taylor expansions generate quantitative bounds.

7.1 Standard Taylor expansions (about 0)

e^x	$1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$
$\sin x$	$x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$
$\cos x$	$1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$
$\ln(1+x)$	$x - \frac{x^2}{2} + \frac{x^3}{3} - \dots \quad (x < 1, \text{ also at } x=1)$
$(1+x)^\alpha$	$1 + \alpha x + \frac{\alpha(\alpha-1)}{2!}x^2 + \dots \quad (x < 1, \alpha \in \mathbb{R})$
$\arctan x$	$x - \frac{x^3}{3} + \frac{x^5}{5} - \dots \quad (x \leq 1)$
$\frac{1}{1-x}$	$1 + x + x^2 + x^3 + \dots \quad (x < 1)$

7.2 Applications

Local approximations. For x small: $\sin x = x + O(x^3)$, $1 - \cos x = \frac{x^2}{2} + O(x^4)$, $\ln(1+x) = x - \frac{x^2}{2} + O(x^3)$. These shortcut limit evaluation.

Quantitative inequality via remainder. Show $\sin x \geq x - \frac{x^3}{6}$ for $x \geq 0$. Taylor: $\sin x = x - \frac{x^3}{6} + R_4(x)$, where $R_4(x) = \frac{\cos \xi}{120}x^5$ for some $\xi \in (0, x)$. (The Taylor polynomials of degree 3 and 4 for $\sin x$ around 0 are identical, since the x^4 coefficient vanishes; using R_4 gives the sharpest standard form.) For $x \in [0, \pi/2]$, $\cos \xi \geq 0$, so $R_4 \geq 0$, giving the inequality immediately. For $x > \pi/2$, $x - x^3/6$ is already very negative while $\sin x \geq -1$, so the inequality holds trivially.

Functional bounds from derivative bounds. If $f'' \geq 0$ on I , then for any $a \in I$:

$$f(x) \geq f(a) + f'(a)(x-a) \quad (\text{tangent below the graph}).$$

This is the tangent-line inequality, a special case of Taylor's theorem with $n=1$ and a sign-controlled remainder.

Showing two functions agree to order n . If f and g have the same Taylor polynomial of degree n at a (i.e., $f^{(k)}(a) = g^{(k)}(a)$ for $k=0, \dots, n$), then $f(x) - g(x) = O((x-a)^{n+1})$. Useful for asymptotic comparisons.

8 Curve sketching — pulling it all together

To sketch $y = f(x)$, gather the following data and assemble:

- Domain, symmetry, periodicity.** Even/odd? Periodic?
- Intercepts and sign.** Where does f cross axes? Where is $f > 0$ or $f < 0$?
- Asymptotes.**
 - Vertical: where $f \rightarrow \pm\infty$ (often at zeros of a denominator).
 - Horizontal: $\lim_{x \rightarrow \pm\infty} f(x) = L$.
 - Oblique: $f(x) = mx + c + o(1)$ as $x \rightarrow \infty$, with $m = \lim_{x \rightarrow \infty} \frac{f(x)}{x}$ and $c = \lim_{x \rightarrow \infty} (f(x) - mx)$.

4. **Monotonicity.** Solve $f'(x) = 0$, find sign of f' on each interval between critical points.
5. **Concavity and inflections.** Solve $f''(x) = 0$, find sign of f'' on each interval.
6. **Extrema.** Apply first or second derivative test at each critical point.

9 Quick-reference cheat sheet

Pattern in the problem	Tool / trick to try first
Show f increasing/decreasing on I	Sign of f' on I
Show $f \leq g$ on I	$h = g - f$, show $h(a) = 0$ and $h' \geq 0$
Find local extrema	Critical points, then 1st- or 2nd-derivative test
$f''(c) = 0$ at a critical point	Higher-derivative test or 1st-derivative test
Find global max/min on $[a, b]$	Compare f at critical points and endpoints
Two variables linked over time	Implicit differentiation in t (related rates)
Estimate small change Δy	$\Delta y \approx f'(x)\Delta x$
Tangent / normal at $(a, f(a))$	Slope $f'(a)$; normal slope $-1/f'(a)$
Tangent to parametric curve	$\dot{y}/\dot{x}$
Tangent to implicit curve $F(x, y) = 0$	$-\frac{F_x}{F_y}$
Approximate root of $f(x) = 0$	Newton: $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$
Sum of reciprocals of polynomial roots	$S_1 = -\frac{p'(0)}{p(0)}$; or Vieta on reversed poly
Higher reciprocal power sums S_k	Differentiate p'/p repeatedly, evaluate at 0
Inequality involving $\ln, e^x, \sin$	Tangent-line family or convexity
Inequality with constraint $\sum x_i = \text{const}$	Jensen on the right convex function
Strange inequality: tries fail	Cascade — iterate “ $h(a) = 0 + h' \geq 0$ ” on derivatives
Bound $ f(x) - P_n(x) $	Lagrange remainder: $ f^{(n+1)} /((n+1)!) x - a ^{n+1}$
Show $f(x) - g(x) = O((x - a)^{n+1})$	Match Taylor coefficients up to order n
Limit involves $\sin x, \ln(1 + x), e^x - 1$	Replace by Taylor series and cancel leading terms
Sketch a curve	Domain, sign, asymptotes, f' , f'' , extrema — in that order
Distance from point to curve	Minimise d^2 , not d
Optimisation under constraint	Reduce to one variable, then differentiate (or Lagrange)

Three universal moves: (1) For inequalities, define $h = \text{LHS} - \text{RHS}$ and chase signs of derivatives. (2) For polynomials with awkward symmetric expressions in roots, reach for the logarithmic derivative or Vieta on the reversed polynomial. (3) For limit and bound problems near a point, expand in Taylor series first — the leading non-vanishing term tells you everything.